
MP Online • Internship Project • 2025-26

GrievanceHub

Digital Grievance Redressal System

Requirements Traceability Matrix (RTM)

Mapping Requirements to Design and Test Cases

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MP Online Internship Project

1. Overview

The RTM maps each business requirement to its corresponding design component, API endpoint, and test case. This ensures complete coverage and verifiability of all stated requirements.

2. Traceability Matrix

Req ID	Requirement	Design Component	API / Module	TC ID	Status
F-01	User Registration	User model, auth_service.py	POST /api/auth/register	TC-01	Pass
F-02	User Login	JWT security.py	POST /api/auth/login	TC-02	Pass
F-03	Anonymous Submission	grievance_service.py	POST /api/grievances/submit	TC-03	Pass
F-04	Logged-in Submission	grievance_service.py, dependencies.py	POST /api/grievances/submit/logged-in	TC-04	Pass
F-05	Ticket ID Generation	ticket_generator.py	Utility function	TC-05	Pass
F-06	Ticket Tracking	ticket_router.py	GET /api/tickets/{ticket_id}	TC-06	Pass
F-07	Citizen Dashboard	Dashboard.jsx, grievance_router.py	GET /api/grievances/my	TC-07	Pass
F-08	Admin Dashboard	AdminDashboard.jsx, admin_router.py	GET /api/admin/grievances	TC-08	Pass
F-09	Status Management	admin_service.py	PATCH /api/admin/grievances/{id}	TC-09	Pass
F-10	Admin Remarks	admin_service.py	PATCH /api/admin/grievances/{id}	TC-10	Pass

3. Component to Requirement Mapping

Component / File	Type	Satisfies Requirements
user_model.py	Backend - DB Model	F-01, F-02
grievance_model.py	Backend - DB Model	F-03, F-04, F-05, F-06, F-07, F-08, F-09, F-10
auth_service.py	Backend - Service	F-01, F-02
grievance_service.py	Backend - Service	F-03, F-04, F-07
admin_service.py	Backend - Service	F-08, F-09, F-10
ticket_generator.py	Backend - Utility	F-05
auth_router.py	Backend - Router	F-01, F-02
grievance_router.py	Backend - Router	F-03, F-04, F-07
ticket_router.py	Backend - Router	F-06
admin_router.py	Backend - Router	F-08, F-09, F-10

Component / File	Type	Satisfies Requirements
security.py + dependencies.py	Backend - Core	F-02 (JWT), F-04, F-07, F-08, F-09, F-10 (Auth guard)
Login.jsx + Register.jsx	Frontend - Pages	F-01, F-02
SubmitGrievance.jsx	Frontend - Pages	F-03, F-04
TrackTicket.jsx	Frontend - Pages	F-06
Dashboard.jsx	Frontend - Pages	F-07
AdminDashboard.jsx + GrievanceTable.jsx	Frontend - Pages	F-08, F-09, F-10